Urban Tree Diversity and Heat Stress: A Neighbourhood-Scale Analysis

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Abstract

Across Europe, cities are placing a greater value on having a higher diversity of trees to address extreme heat variations. However, figuring out how this diversity helps local temperatures is not always straightforward. This study analysed Strasbourg's urban trees, matching records for over 82 000 trees with satellite temperature from 2022 to 2024 (n = 15 neighbourhoods). Well known diversity measures were applied such as the 10-20-30 guideline to display diversity and assess ecological vulnerability in all neighbourhoods.

To test whether diversity is associated with temperature directly or through structure on a neighbourhood scale, a multivariate format was employed. In the first model (diversity only), a positive correlation between tree diversity and land surface temperature ($\beta = 1.21$) was revealed. At a 0.05 threshold this result was not statistically significant, partly due to the small sample size (n = 15). When accounting for canopy, concrete, and tree density in the second model, the diversity coefficient climbed to +2.09 and the model fit increased substantially ($R^2 = 0.023$ to 0.913) with impervious surfaces emerging as the strongest factor.

These results suggest that in Strasbourg, urban forest diversity tracks urban heat stress, although this relationship is sensitive to individual neighbourhoods due to small sample size. Overall, diversity coincided with higher land surface temperature, consistent with municipal strategies that prioritize diversity to overcome the heat in warmer districts, which could induce positive correlation at a neighbourhood scale. (Plan Canopée, 2020)

1. Introduction

Cities are becoming denser and hotter, making urban heat an increasingly serious environmental and public health concern. Rising summer temperatures intensify heat stress, increase mortality risk, and reduce urban safety (Kovats & Hajat, 2008). Urban trees primarily help through shading and evapotranspiration , and are central to many climate adaptation strategies.

Beyond tree presence and canopy cover, tree diversity has become a key component for urban forest adaptation. To reduce ecological vulnerability, many cities apply the **10-20-30 rule**, which limits the dominance of any single species, genus, or family within the urban canopy (Santamour, 1990). The rule is designed to prevent large-scale forest loss from pests, diseases, or climate stress.

The 10-20-30 is intended to reduce long-term vulnerability and prevent forest collapse (Freiburger et al., 2025). Instances where high vulnerability scores coincide with high diversity values should not be interpreted as a temperature effect. This distinction is crucial, because it separates **resilience** (keeping the urban forest alive) from **performance** (the physical cooling provided by the trees). In the context of thermal stress, the 10-20-30 rule ensures that the cooling benefits provided by tree structure are not lost to future pests or climate stress.

Shannon and **Simpson** indices have been used to describe ecological variety of communities, offering complementary and effective indices to determine biodiversity . In this study, they will be used for comparison and correlation with LST, however, they should also be used as tools to measure unequal diversity in Strasbourg's neighbourhoods and help urban planning. It is important to note that, whether such diversity influences urban thermal conditions is uncertain even though the local mechanisms are relatively well documented (Li et al., 2024). These effects are harder to verify in complex urban environments where multiple factors come into play (Chinchilla et al., 2021; Mishra et al., 2015). However, this relationship remains well less documented at the neighbourhood scale.

In this study I investigated these questions in Strasbourg by combining municipal tree inventory with satellite-derived summer land surface temperature data. Examining tree diversity, species dominance, and thermal conditions across the city. By analysing ecological vulnerability indicators and surface temperature patterns, this study aims to clarify the relationship between urban forest composition and urban heat. This study has 3 objectives :

Identify fragile neighbourhoods with the 10-20-30 diversity rule. Map where the city is hottest by measuring summer land surface temperature across Strasbourg and test if diverse neighbourhoods experience changes in temperatures, before and after accounting for structural characteristics.

2. Methods

2.1 Study Area and Analyses

The analysis was conducted in Strasbourg, focusing on the 15 neighbourhoods delimiting the metropolis of Strasbourg. The study adhered to the 15 districts defined during the inventory compilation rather than standard administrative boundaries. Using this specific scale aligns with how the data was collected. This scale provided enough trees in each area sufficient for explanatory analysis while keeping the diversity and temperature readings consistent across the whole city. Note

that not all trees of the metropolis of Strasbourg were included in the dataset, such as dense forests, trees on private property, and trees under communal jurisdiction outside of Strasbourg.

All geospatial processing, including municipal tree inventory, LST computation, and zonal statistics for impervious cover, was performed within **Google Earth Engine (GEE)** (Gorelick, N., et al., 2017). By using the GEE API and cloud data, the LST calculation was heavily facilitated as it helped overcome the huge size of Landsat images (U.S. Geological Survey, 2022) . Final cartographic maps were produced using **QGIS** (QGIS development team, 2024). Lastly, the statistical tests were performed using python. (Python Software Foundation, 2023).

2.2 Tree Data and Data Cleaning

The main dataset used was the Municipal Tree Inventory, containing **82 387** located trees across Strasbourg, the species and genus were also provided for every tree (except a small minority) (*Patrimoine arboré*, n.d.). To organise the data, a two step process was devised. Separating genus from species to create a clean table and then assigning families to every genus using a manually validated table covering 96 genera.

While the final dataset does contain the spatial data for every tree, about 385 of them (less than 0.5%) did not have enough information to be assigned to a family. Therefore, they were excluded from biodiversity and dominance calculations to ensure reliability.

2.3 Tree Diversity Metrics

To evaluate the diversity of Strasbourg's urban forest within neighborhoods, species richness (S) Shannon (H') and Simpson (1 - D) indices were calculated (Shannon, 1948; Simpson, 1949). These indices help understand how many different species are in a given area (S), whether the system (neighbourhood) has many species and how evenly distributed they are (H'), and lastly to check if a neighbourhood is being dominated by 1 or a few species (1 - D).

Furthermore, the neighbourhoods were evaluated with the 10-20-30 rule. This rule is a better indicator for **urban management** and sets a threshold of no more than 10% of a single species, 20% of a genus, and 30% of a family. The Risk score (0-3) counts how many thresholds a neighbourhood exceeds. For context, one dominant type of tree means they would all be vulnerable to the same pest or disease.

0: Passes all thresholds = diverse and resilient.
1: One threshold exceeded = low vulnerability.
2: Two thresholds exceeded = moderate risk.
3: All thresholds exceeded = high risk. The neighbourhood relies on one dominant tree type across all levels.

Higher scores indicate **greater vulnerability** which means neighbourhoods could suddenly lose cooling benefits if their dominant trees fail to survive ecological shock.

2.4 Land Surface Temperature (LST)

Land Surface Temperature (LST) data came from **Landsat 8** and **9** satellites from June to September across 2022-2024. Three years were used to ensure that typical heat conditions were captured rather than a single unusual summer.

The image was processed in four steps. Firstly, eliminating pixels covered by clouds or shadows then transforming satellite values into temperatures (°C) and creating the median value across all three summers to smooth out weather anomalies. Lastly, the mean temperature was computed for each neighbourhood separately.

This computed a single, stable temperature value per neighbourhood representing typical summer heat exposure.

2.5 Impervious Surface Cover

To account for grey infrastructure, the average **impervious surface** cover for each neighbourhood was calculated using the ESA WorldCover v200 (2021) dataset (European Space Agency, 2021). The analysis focused on Class 50 ("Built-up") at a 10 m resolution, which includes roads, buildings, and artificial surfaces. To keep spatial resolution consistent with Strasbourg's local coordinates, zonal mean reduction was run in the UTM Zone 32N (EPSG:32632) projection. This allowed for the conversion of land cover pixels into a clean percentage of neighbourhood area classified as built-up in relation to neighbourhood proportion, for every district in Strasbourg.

Furthermore, ESA WorldCover (2021) was used over "Copernicus HRL Imperviousness (2018)" due to lower resolution scale, data being too old compared to the study period and its incompatibility with Google Earth Engine. However, a substantial benefit was lost as it offered continuous 0-100% impervious values.

2.6 Multivariate Framework

LST was tested whether it was directly correlated to tree diversity or indirectly through tree structure and characteristics. To do this, a two-step process was run across Strasbourg's districts. The goal was to determine if diversity is a standalone cooling factor or if it is also influenced by biophysical cooling mechanisms and density. By splitting the analysis, **underlying characteristics** and their **independent importance** could be uncovered**.** Distance to water was initially analyzed as a control variable, but was removed from the final model as it did not meaningfully change the coefficient or model fit and further constricted the degrees of freedom.

Model 1 :

$LST = \beta_0 + \beta_1(Shannon) + \varepsilon$

Estimates the correlation between taxonomic diversity and temperature without controls.

Model 2:

$LST = \beta_0 + \beta_1(Shannon) + \beta_2(Density) + \beta_3(CrownArea) + \beta_4(Impervious) + \varepsilon$

Estimates the effect of diversity while controlling for tree density (trees/ha), mean crown area (m^2), and percentage of impervious surface. Mean crown area was part of the original dataset whereas tree density was calculated for each neighbourhood.

The standard error was also calculated, mostly for statistical tests performed afterwards.

2.7 Statistical Significance and Diagnostics tests

To verify the significance of the results from the multivariate analysis, three different statistical tests were conducted.

A **T-test**, evaluated the significance of individual regression coefficients, in other words if a specific variable (β_1(Shannon), β_3(CrownArea) etc.), has a statistically reliable effect on LST, or if it is due to random chance. The t-statistic value measures how many standard errors the coefficient is away from 0.

Secondly, the **Clogg test** (also known as the z-test) was used to assess if the change in coefficient between models was statistically significant. The Clogg test determines if this change is larger than expected from sampling variation alone.

Lastly, the **F-test** was used to evaluate whether adding variables helped with model fit. It helps determine if additional variables collectively explained significantly more variance than by random variation.

Due to the small sample size (n = 15) several diagnostics procedures were run. Model fit was evaluated using R^2 and **adjusted R^2** due to instability produced by the 4 variables. **Cook's distance** was measured to identify if certain neighbourhoods are more influential than others in terms of individual coefficients. If the value from Cook's distance is over 4/n (0.267), that means that the neighbourhoods influence and pulls the coefficient towards them. The **Leave-One-Out** (LOO) method was also used, running the regression analysis every time removing 1 neighbourhood to see if the results were fragile and unstable. A larger range would mean that they are unstable, and the coefficients depend on the neighbourhoods used.

3. Results

3.1 Tree Diversity Patterns

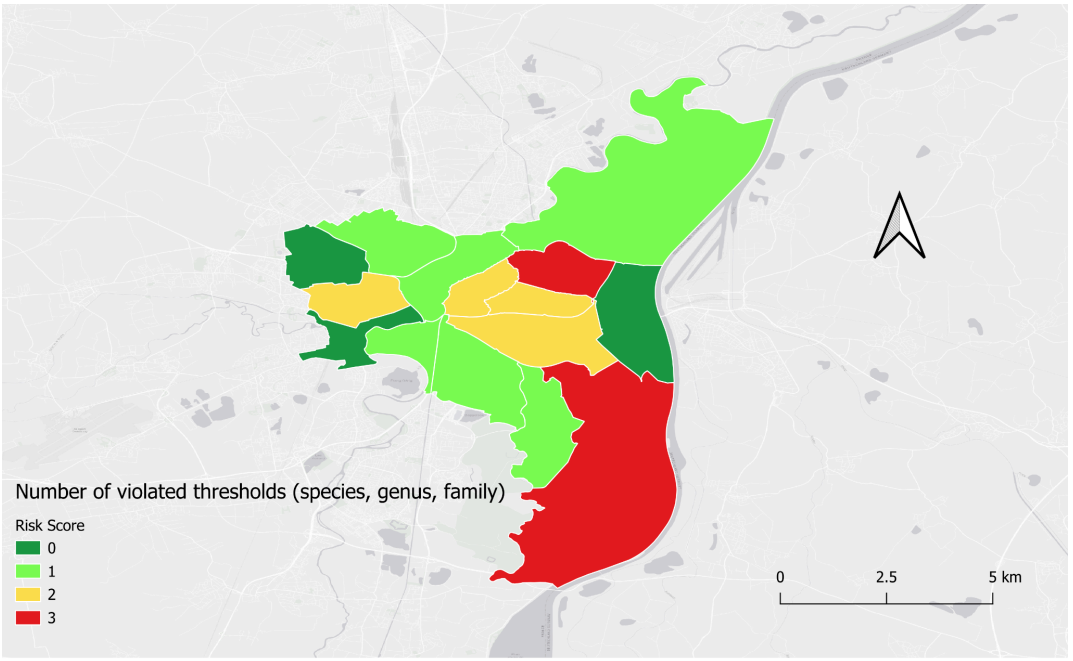
In Strasbourg, some neighbourhoods have few specific species while some have more heterogeneous pockets where the Shannon and Simpson values spike. All diversity indices are internally consistent with zero missing values. All values are available in the appendix including, Shannon, Simpson, richness, Top species/genus/family, Highest percentage of species/genus/family, risk score and mean LST, all **for each neighbourhood**. (See Annexe A)

3.2 Taxonomic Dominance and Risk

Application of the 10-20-30 rule across every neighbourhood found that some areas fully comply (risk score = 0), while others exceed one or multiple thresholds. A small number reach the maximum risk score.

These variations are visible in the map below (see Figure 1).

Figure 1 - Urban tree diversity risk (10-20-30 rule).



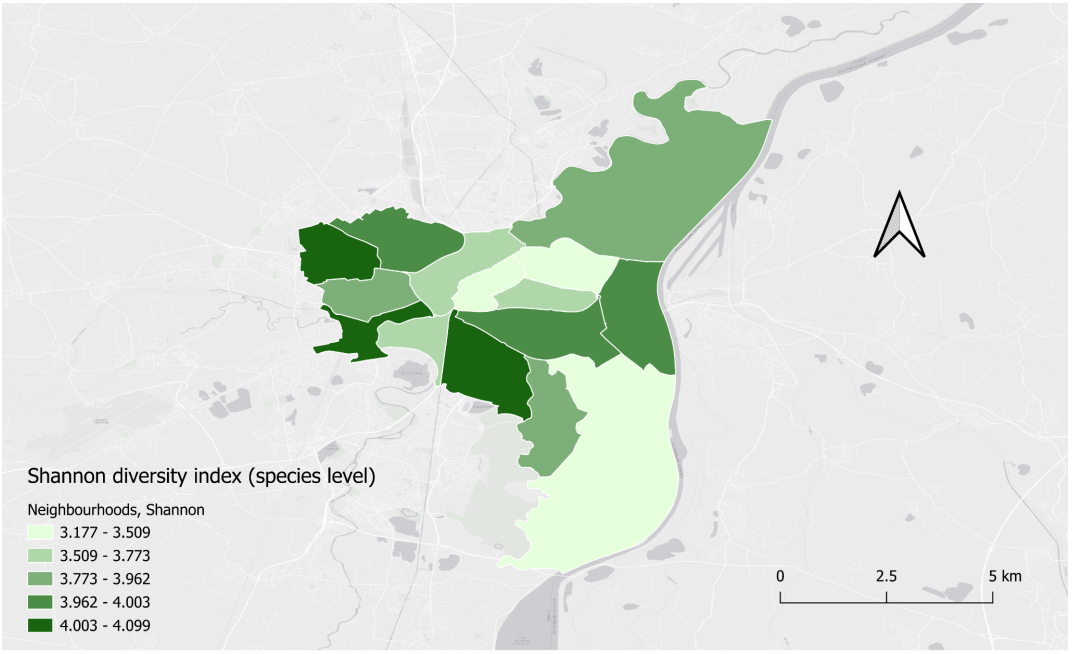
Data: Strasbourg tree inventory (Eurométropole)
Processing: Google Earth Engine

Green indicates neighbourhoods with full compliance (score 0), light green (score 1), orange (score 2), and red (score 3) indicate increasing vulnerability due to dominance of a specific taxonomic group.

Independently, a **variety** of risk scores is present through the metropolis of Strasbourg, with three neighbourhoods passing the 10-20-30 rule demonstrating very low vulnerability to ecological shocks. Six neighbourhoods exceed one threshold which could be perceived as low ecological risk. In addition, four exceed two thresholds and two exceed three showing that there are still neighbourhoods that show ecological vulnerability and should be placed under caution. In all, most neighbourhoods in Strasbourg show low ecological vulnerability, while some may need intervention.

3.3 Shannon Index and LST

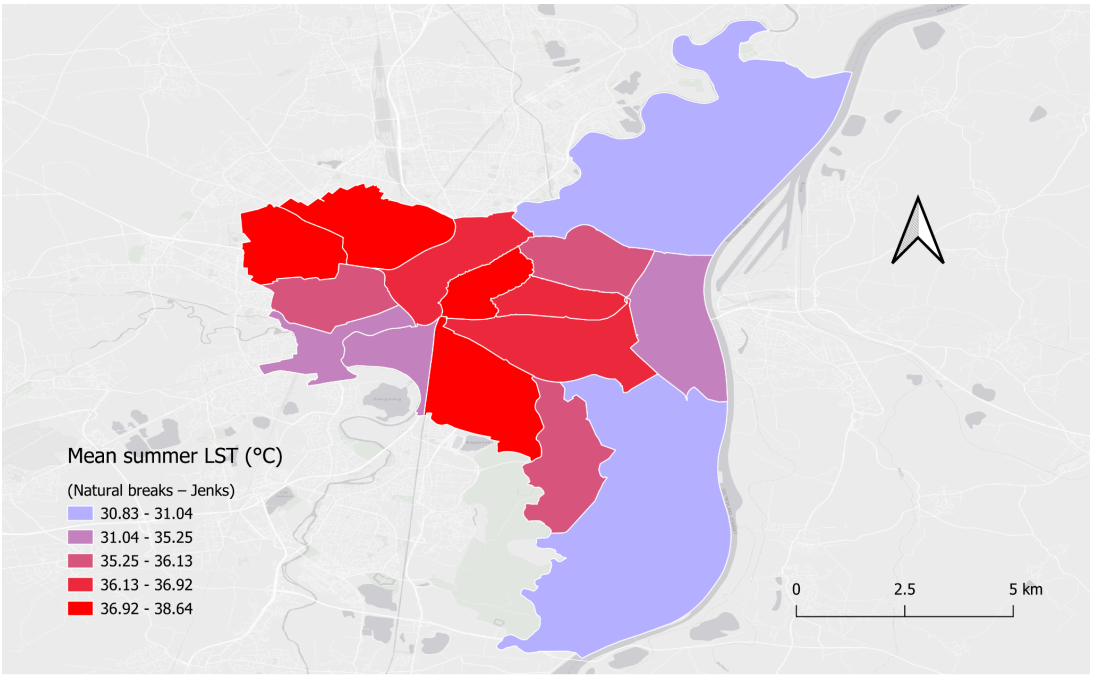
Figure 2 - Tree species diversity (Shannon index) per neighbourhood.



Data: Strasbourg tree inventory (Eurométropole)
Processing: Google Earth Engine

Darker green tones indicate neighbourhoods with higher species diversity, lighter shades reveal areas with lower variety.

Figure 3 - Mean summer land surface temperature (LST) per neighbourhood.

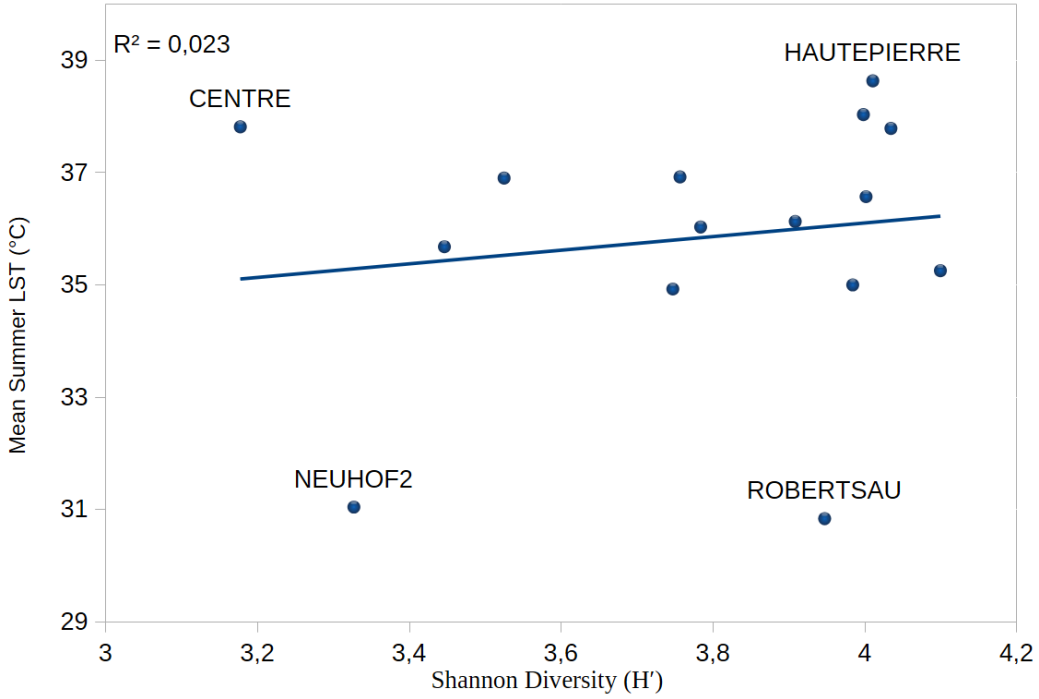


Data: Landsat 8–9 (USGS), Strasbourg Eurometropolis
Processing: Google Earth Engine

Warmer tones (reds/orange) indicate neighbourhoods with higher surface temperatures, while cooler tones (blue/purple) reveal areas with lower surface temperatures. Natural breaks - Jenks was chosen because it maximizes the differences between classes, and minimizes differences within classes.

When comparing these 2 maps, the Shannon index and mean LST are visually revealed and do not follow the understandable presumption that both go **hand-in-hand**. Some neighbourhoods that have the least diversity show the highest average LST and vice-versa (see figure 4).

Figure 4 - Relationship between Shannon Diversity and Mean Summer LST (°C) across Strasbourg neighbourhoods.



This scatterplot demonstrates the weak relationship, where several neighbourhoods deviate from the general trend. 4 specific cases are highlighted below (see table 1).

Table 1: Contrasting cases illustrating the absence of consistent relationships between Shannon and LST.

Neighbourhood	Shannon	LST	Note
Hautepierre	4.01	38.64	High diversity, hottest
Neuhof2	3.33	31.04	Low diversity, cool
Robertsau	3.94	30.83	High diversity, cool
Centre	3.18	37.80	low diversity, hot

Hautepierre, a neighbourhood with a very **high Shannon index** (4.01) ends up being the **hottest** neighbourhood with a mean temperature across 3 years and through the month of June to September of 38.64 °C. Furthermore, The Neuhof2 with the second **lowest diversity** also ends up being the neighbourhood with the **lowest mean temperature** across 3 years and through the month of June to September of 31.04 °C.

However, this relationship between high temperature and high diversity is **not homogenous**, as illustrated in Robertsau where the diversity index is very high (3.94) and the temperature is the coldest out of any neighbourhood. Centre is another contradicting case where the diversity is very low however it is one of the warmest. These four contrasting neighbourhoods demonstrate the unevenness between temperature and diversity.

Regarding this confounding link between diversity and temperature, it is also visible that diversity in Strasbourg is disproportionately spread out. Ranging from 3.177 to 4.09. Shannon diversity was relatively high across Strasbourg neighbourhoods ($H' = 3.177 - 4.09$) compared with values reported for street-tree inventories in other cities. (Galle et al., 2021)

LST has a surprisingly large range too depending on the neighbourhood between 30.83 and 38.64 °C. This range can be explained by the Urban Heat Island (UHI) effect and other variables. Where Urban centers are remarkably hotter than surrounding neighbourhoods.

3.4 Multivariate Analysis

To see if heat and diversity was correlated a multivariate framework was used to investigate the characteristics and structure that influences the correlation between heat and diversity.

Here below is a table recapitulating the results regarding the multivariate analysis. The full table including the standard deviation is available in the annexe (see Annexe B)

Table 2: Total and direct effects of tree diversity on Land Surface Temperature

	Model 1: Total effect	**Model 2: Direct effect**
Predictor	**β**	**β**
Intercept	31.25	21.34
Shannon diversity	**+1.21**	**+2.09**

Tree density (tree/ha)	-	+0.02
Mean crown area (m²)	-	+0.001
Impervious surface (%)	-	**+0.111**
Model fit	-	-
R^2	0.023	0.913
n	15	15

Model 1 estimated a +1.21 °C change in LST per unit increase in Shannon diversity. The second model, took into account other variables such as tree density, mean crown area and impervious surface estimated at 2.09. Furthermore, it detected an increase in model fit of **0.89** indicating that the second model is a better predictor of temperature.

3.5 Statistical tests

To test if the results were significant 3 different statistical tests were run and it is worth highlighting that all tests were done at a 5% significance level.

The T-test was used to see if each individual coefficient (β) was reliably significant in defining if diversity really does correlate with higher LST. The findings are listed in the table below (see Table 3)

Table 3 T-tests for individual coefficient significance

Variable	Coefficient	t-statistic	p-value	Result
Shannon (Model 1)	+1.21	0.55	0.59	**Not significant**
Shannon (Model 2)	+2.09	2.75	0.020	**Significant**
Impervious (Model 2)	+0.111	7.96	<0.001	**Highly significant**

The T-test indicates that in the first model Shannon is insignificant and therefore (diversity) can not be interpreted and does not predict temperature because the p-value (0.59) is over 0.05. However, in the

second model where tree structure was accounted for, Shannon was significant (p-value < 0.05) but coefficients should be interpreted cautiously given the small sample size. Impervious cover in the second model was also highly significant, with a p-value of less than 0.001, highlighting that concrete and built surfaces are a major heat driver.

In addition, to see if the coefficient increase from 1.21 to 2.09 was significant or not, we performed the Clogg test.

Table 4 - Clogg test results for coefficient comparison between models.

Comparison	difference	Z-statistic	p-value	Result
Shannon : Model 1 → 2	1.21 → 2.09	0.38	**0.7**	**Not significant**

The p-value for the Clogg test came back as 0.7 signaling that the increase of coefficient is insignificant meaning that the coefficient increase could be due to chance.

Lastly, an F-test was used to determine if Model 2 and more precisely its structural variables, improved the model.

Table 5 - F-test results for model improvement

Metric	Value	Result
R^2 increase	2.3% → 91.3%	Massive improvement
F-statistic	**34.12**	**Huge increase**
p-value	0.000015	$p < 0.001$

The F-value returned as 34.12 meaning that when adjusting with structural characteristics the "model fit" greatly improves. The p-value came back as lower than 0.001 indicating that the improvement is significant and interpretable.

Regarding diagnostics the adjusted R^2 in the second model retained a very high value ($R^2 = 0.865$), suggesting strong explanatory competency despite the small sample. Lastly, the Cook's distance and

LOO effect displayed surprising and important results. They complement each other by providing the overall change (Cook's) and the specific coefficient change (LOO).

Table 6 - Key neighbourhoods and influence on diagnostics

Neighbourhood	Cook's distance	LOO Shannon β
Robertsau	1.424 (extreme)	=4.04 (+93%)
Neuhof2	0.641 (high)	≈1.5 (-28%)
Centre	0.385 (moderate)	≈2.5 (+20%)

Threshold for influence 4/n = 0.267

These results support the visual comparison of the absence of consistent relationship between diversity temperature. Robertsau violates the Cook's threshold by over 4 times, demonstrating its major effect, and once removed from the regression analysis the β value for diversity index augments 93% indicating that Robertsau was substantially lowering the diversity to temperature correlation. Neuhof violates the threshold too with a Cook's distance of 0.641. When removing Neuhof2 from the regression analysis **β Shannon** lowers from 2.09 to 1.5 indicating that Neuhof2 was doing the opposite of Robertsau, suggesting that high diversity correlated with high temperatures. Lastly, Centre exceeded the threshold minimally by 0.118 showing minimal effects compared to the two other neighbourhoods. In addition, the LOO value for Shannon came back as 2.5 meaning that Centre was lowering **β Shannon** singlehandedly and at a significant effect.

4. Discussion

This study revealed a **counter-intuitive finding**, tree diversity correlates positively with Land Surface Temperature (β = +1.21). However, it is crucial to point out that this relationship only becomes a significant predictor after accounting for structural factors, and can not confirm if the increase of coefficient represents a true strengthening of the relationship or improved precision. When controlled for grey infrastructure and tree characteristics the diversity coefficient augmented to +2.09 (73% increase). The nearly 90% jump in model fit (R^2) suggests that the built environment was previously obscuring the real relationship between diversity and heat.

Diagnostic analyses highlight that individual neighbourhoods sway the results making the relationship between temperature and diversity **sensitive**. Cook's distance and LOO tests reveal that some neighbourhoods are disproportionately influencing the regression coefficients, indicating that results

should be interpreted as **explanatory**. This unreliability comes from both the small sample size (n=15) and the heterogeneity throughout Strasbourg. This instability reinforces that diversity and temperature do not follow a consistent pattern through the neighbourhoods

This study found that diversity was not acting as a physical cooling agent but instead appeared to track urban heat. This can be explained by **municipal policies** (Plan Canopée, 2020), which in Strasbourg have prioritized thermally stressed districts for diversification as a resilience strategy. Furthermore, this plan has started in 2020 and has already planted around 5500 trees, which would augment diversity, but since the trees are very young, they have yet to make a significant impact in terms of evapotranspiration and canopy cover (Plan Canopée, 2020). One plausible explanation would be that diversity is not the driver of heat at this scale, and instead reflects municipal strategies to aid long-term in thermally stressed areas. Testing this theory would require information on planting data.

At this scale, urban composition overwhelms vegetation characteristics. Impervious surfaces emerged as the **dominant** driver ($\beta = +0.111$), explaining the vast majority of thermal variance. In contrast, tree density and crown area showed negligible effects ($\beta_{crown} \simeq +0.001$). This suggests that in high density environments, even large crowns can not overcome the substantial effect of impervious cover. However, the expectation would be for the sign to be negative, as multiple studies have shown how canopy decreases temperature through several processes. A plausible interpretation is that crown area (and diversity) is clustered in hot and dense districts, targeted by **planting policies**, which could make positive coefficients at this scale.

Analyzing at the neighbourhood level further constrains the ability to interpret the relationship between temperature and diversity. At smaller scales, diversity coincides with cooler microclimates because it co-varies with structural and functional complexity (Cai et al., 2022). At the scale of entire urban districts, localized effects are **diminished** by land use and built structures. In addition, satellite-derived surface temperature integrates thermal signals from roofs, roads, and other impervious elements, reducing the visibility of vegetation related effects.

Future studies should consider incorporating more characteristics like tree size, maturity, total leaf area index, or unique capabilities of certain tree species, to increase precision.

5. Conclusion

This study provides a **reproducible** assessment of Strasbourg’s urban forest by combining municipal tree inventory data with satellite-derived thermal imagery to evaluate biodiversity and heat exposure across the city's neighbourhoods. The analysis identified neighbourhood-level ecological

vulnerability, map thermal hotspots, and evaluate the cooling effects of tree diversity has successfully been done through taxonomic, structural, and spatial analyses.

The central finding is the counter-intuitive positive correlation between diversity and Land Surface Temperature. This is possibly explained by municipal policies which aim to offset high neighbourhood temperatures. Impervious surface was strongly associated with LST and was the strongest predictor in Model 2. This highlights the need for **ecology oriented renovation** in neighbourhoods with a lot of impervious surface.

Ultimately, this study suggests effective urban forestry requires a two-way strategy. On one hand, you need high taxonomic diversity to act as an **insurance policy** against future pests and climate collapse. On the other hand, **high canopy density** and **reduced concrete** are needed for thermal relief today.

Data and code availability: The municipal tree inventory data is available in the Eurométropole de Strasbourg open data portal updated every month. All python and GEE processing scripts are available on GitHub at: https://github.com/AlexJMilner/strasbourg-urban-heat-analysis

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Appendix

Appendix A : A complete inventory of Tree Biodiversity per neighbourhood in Strasbourg

Quartier	Total arbres	Richness	Shannon	Simpson	Dominant Species	% Species	Dominant Genre	% Genre	Dominant Family	% Family	LST average (°C)	Risk scores
Neuhof 2	1 868	81	3.32	0.94	Acer pseudoplatanus	12.1	Acer	23.9	Sapindaceae	33.8	31.0	3
Centre	1 833	110	3.17	0.91	Platanus acerifolia	18.1	Tilia	25.0	Malvaceae	25.0	37.8	2
Elsau	1 927	121	3.74	0.95	Fraxinus excelsior	10.1	Acer	14.7	Sapindaceae	18.1	34.9	1
Port du Rhin	2 004	125	3.98	0.96	Quercus cerris	9.2	Quercus	14.7	Rosaceae	17.1	34.9	0
Bourse–Esplanade –Krutenau	2 939	139	3.52	0.94	Platanus acerifolia	17.9	Acer	20.0	Sapindaceae	25.2	36.9	2

Koenigshoffen	2 787	154	3.90	0.96	Acer pseudoplatanus	11.0	Acer	20.4	Sapindaceae	21.2	36.1	2
Neuhof 1	3 874	154	3.78	0.95	Platanus acerifolia	12.2	Acer	16.3	Sapindaceae	17.8	36.0	1
Tribunal–Gare–Porte de Schirmeck	4 216	160	3.75	0.95	Platanus acerifolia	11.1	Acer	13.7	Sapindaceae	23.4	36.9	1
Hautepierre	5 006	174	4.01	0.96	Carpinus betulus	9.4	Acer	16.9	Sapindaceae	19.5	38.6	0
Montagne-Verte	2 369	176	4.09	0.96	Tilia euchlora	7.2	Acer	14.1	Rosaceae	19.1	35.2	0
Meinau	5 816	195	4.03	0.96	Platanus acerifolia	10.8	Tilia	17.4	Malvaceae	17.4	37.7	1
Cronenbourg	5 136	202	3.99	0.96	Platanus acerifolia	12.3	Tilia	14.3	Sapindaceae	16.1	38.0	1
Neudorf	5 497	207	4.00	0.96	Platanus acerifolia	11.5	Tilia	22.4	Malvaceae	22.4	36.5	2

Orangerie–Conseil des XV	8 331	218	3.44	0.93	Acer pseudoplatanus	13.2	Acer	23.9	Sapindaceae	32.0	35.6	3
Robertsau	9 467	229	3.94	0.96	Platanus acerifolia	12.5	Acer	15.4	Sapindaceae	17.0	30.8	1

Appendix B - Total and direct effects of tree diversity on Land Surface Temperature including standard deviation

Model 1: Total effect			**Model 2: Direct effect**	
Predictor	**β**	SE	**β**	SE
Intercept	31.25	8.3	21.34	3.03
Shannon diversity	**+1.21**	2.19	**+2.09**	0.76

Tree density (tree/ha)	-	-	+0.02	0.040
Mean crown area (m²)	-	-	+0.001	0.001
Impervious surface (%)	-	-	**+0.111**	0.014
Model fit	-	-	-	-
R^2	0.023	-	0.913	-
n	15	-	15	-